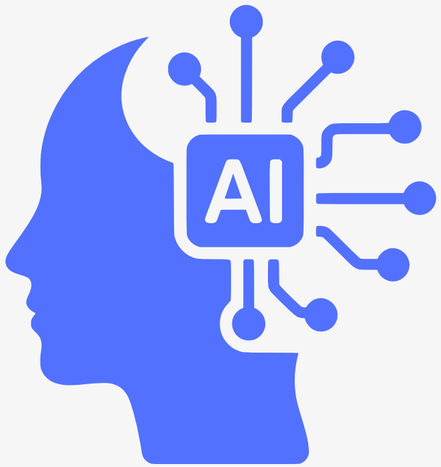
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VECTOR-LESS RETRIEVAL SYSTEMS

NOT EVERY AI PROBLEM NEEDS A VECTOR DATABASE. SOMETIMES, THE BEST RETRIEVAL SYSTEM IS SIMPLY THE DATA YOU ALREADY HAVE.

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WHAT IS VECTORLESS RETRIEVAL?

- Retrieves data without vector embeddings
- Uses structured or relational data
- Works well with exact relationships
- Can use SQL queries and filters
- Often faster and more predictable for structured data

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WHEN DO VECTORS FAIL?

- Exact matching is difficult
- Structured relationships get lost
- Numbers and filters can be unreliable
- Metadata filtering can become complex
- Results may be semantically similar but logically wrong

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RELATIONAL RETRIEVAL

- Retrieves information from relational databases
- Uses tables and relationships
- Uses SQL queries
- Excellent for structured business data
- Precise and deterministic.

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STRUCTURED CONTEXT SYSTEMS

- Give AI structured information instead of raw documents
- Context can come from SQL, APIs, JSON, or application data
- AI receives only relevant information
- Reduces unnecessary context
- Improves accuracy and control

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VECTOR SEARCH VS VECTORLESS RETRIEVAL

Vector Search:

- Semantic similarity
- Embeddings
- Great for unstructured data
- Documents, knowledge bases
- Probabilistic

Vectorless Retrieval:

- Exact/structured retrieval
- SQL/filters/relationships
- Great for structured data
- Users, orders, analytics
- More deterministic

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THE REAL PRODUCTION APPROACH

- Don't choose vectors for everything
- Don't choose SQL for everything
- Use the right retrieval method for the data
- Combine multiple retrieval strategies when needed
- Optimize for accuracy, cost, and control